School of Future Tech

Case Study Report

on

HANGMAN MULTIPLAYER (WORD GUESSING PLATFORM)

by

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1. Introduction to the Case Study

The Hangman Multiplayer Word Guessing Platform is a web based game designed to provide an interactive and engaging experience for users. The project is based on the traditional Hangman game where players try to guess a hidden word by suggesting letters within a limited number of attempts. The purpose of this project is to transform the classic single player Hangman game into a modern web application that includes multiplayer interaction, score tracking, and a community discussion forum.

The application is developed using fundamental web technologies including HTML, CSS, and JavaScript. HTML is used to create the structure of the webpage, CSS is used to design the user interface and improve visual appearance, and JavaScript is used to implement the core logic and functionality of the game.

The platform allows users to guess letters to reveal a hidden word while a hangman figure is drawn step by step when incorrect guesses are made. Additional features such as a leaderboard and forum section increase user engagement by allowing players to track scores and share strategies.

This project demonstrates how simple browser technologies can be used to develop interactive web applications without the need for server side programming.

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2. Problem Statement / Case Background (Abstract)

Traditional Hangman games are usually limited to single player mode and lack features that encourage long term user engagement. Players often have no way to track their performance or compete with others. In addition, most versions of the game do not provide a platform for users to interact with each other or discuss strategies.

The objective of this project is to develop a responsive web application that enhances the traditional Hangman game by adding modern features. These features include multiplayer gameplay, score tracking through a leaderboard, and a forum section for discussion among players.

The system allows players to guess letters from the alphabet to reveal a hidden word within a limited number of attempts. Each incorrect guess results in a progressive drawing of the hangman figure. The application automatically detects win or loss conditions and updates the player score accordingly.

By introducing multiplayer interaction and performance tracking, the platform becomes more engaging and competitive. The forum feature further enhances the experience by enabling players to discuss tricky words and share game strategies.

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3. Problem Statement / Case Study Design

The Hangman Multiplayer platform is designed with a clear and organized user interface that ensures easy navigation and usability. The application is divided into several main sections which help users interact with different parts of the system.

1. The Home section introduces the application and provides navigation links to other parts of the platform.

2. The Play section is the main gameplay area where players attempt to guess the hidden word. The word display initially shows blank spaces representing each letter of the word.

3. An alphabet button grid is provided so that users can select letters. When a correct letter is chosen, it appears in the correct position in the word. If the guess is incorrect, the hangman drawing progresses.

4. The Leaderboard section displays player scores in a structured table format. This encourages players to compete and track their performance.

5. The Forum section allows players to post comments and discuss strategies or difficult words. This feature helps create a small community around the game.

The design focuses on clarity, simplicity, and responsiveness so that the platform can be used comfortably on different screen sizes.

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4. Methods and Algorithms Technology Applied in the Problem Statement / Case Study

The development of the Hangman Multiplayer platform involves the use of several web technologies and logical algorithms.

HTML is used to build the structural layout of the application. It defines elements such as the navigation menu, word display area, alphabet buttons, hangman drawing area, leaderboard table, and forum comment section.

CSS is used to style the user interface and create a visually appealing layout. It controls colors, spacing, button styles, and text formatting. CSS also ensures that the application layout adapts to different screen sizes, making the design responsive.

JavaScript is used to implement the main functionality and game logic. Several important algorithms are used in the system.

1. Random word selection algorithm chooses a word from a predefined list each time the game starts.

2. Letter validation logic checks whether the selected letter exists in the hidden word.

3. Incorrect guess tracking counts the number of wrong attempts made by the player.

4. Hangman drawing progression updates the visual drawing of the hangman figure step by step for each incorrect guess.

5. Win detection algorithm determines when all letters in the word have been correctly guessed.

6. Loss detection algorithm determines when the maximum number of incorrect guesses has been reached.

7. Multiplayer turn switching allows two players to alternate roles such as word setter and guesser.

8. Local storage technology is used to store leaderboard scores and forum comments directly in the browser.

These technologies and algorithms work together to provide a smooth and interactive gameplay experience.

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5. Problem Statement / Case Study Implementation Details and Snapshots

The Hangman Multiplayer application is implemented using three main files that work together to create the complete system.

1. The index.html file contains the main structure of the webpage. It defines the layout of the game interface, navigation menu, leaderboard section, and forum area.

2. The style.css file contains the styling rules used to design the visual appearance of the platform. It controls elements such as button styles, layout alignment, color schemes, and responsive behavior.

3. The script.js file contains the JavaScript code responsible for handling the game logic. It manages word selection, user input handling, hangman drawing updates, score tracking, and multiplayer functionality.

Key features implemented in the system include the following.

1. Dynamic word display where hidden letters appear when guessed correctly.

2. Interactive alphabet buttons that become disabled after being selected.

3. Progressive hangman drawing that visually represents incorrect guesses.

4. Automatic win and loss detection based on player guesses.

5. Leaderboard system that stores and displays player scores.

6. Forum section where players can post and view comments.

In this section of the report, screenshots of the following screens can be added.

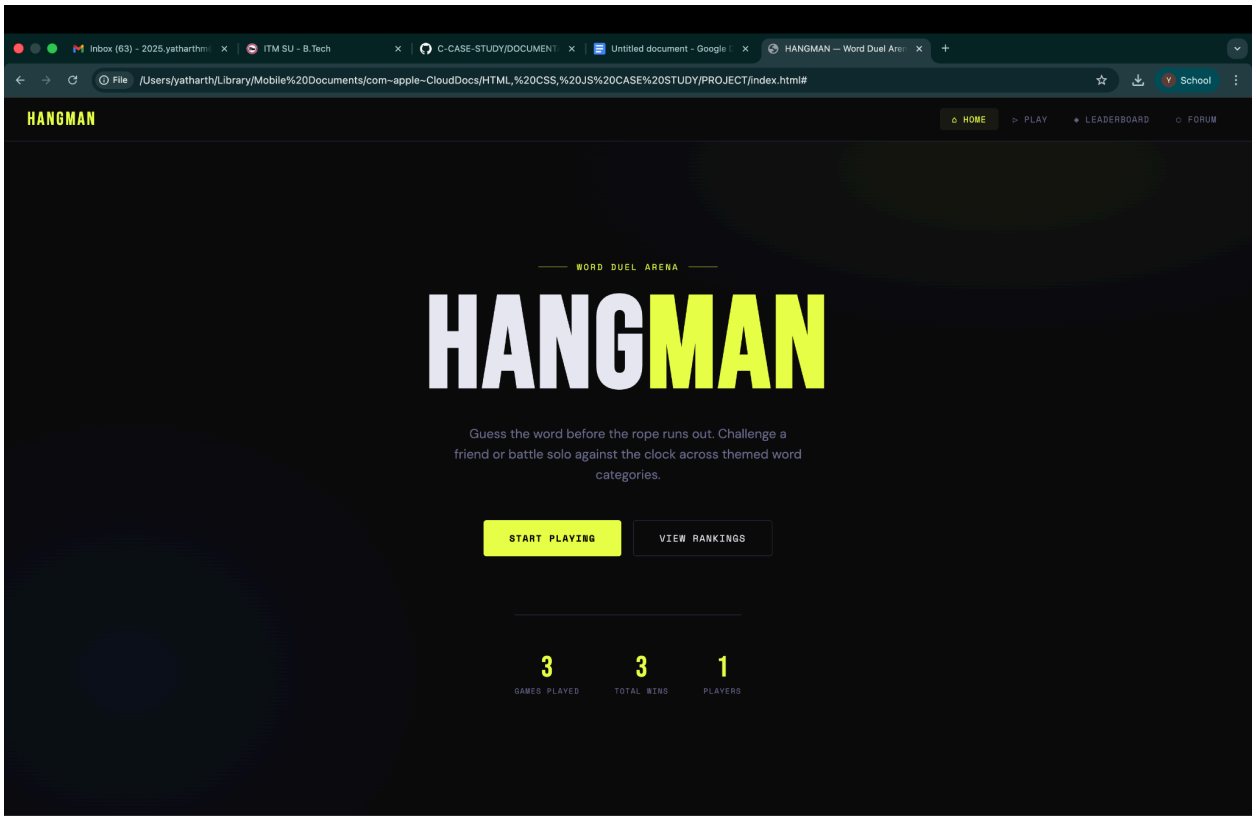
1. Main home page of the application

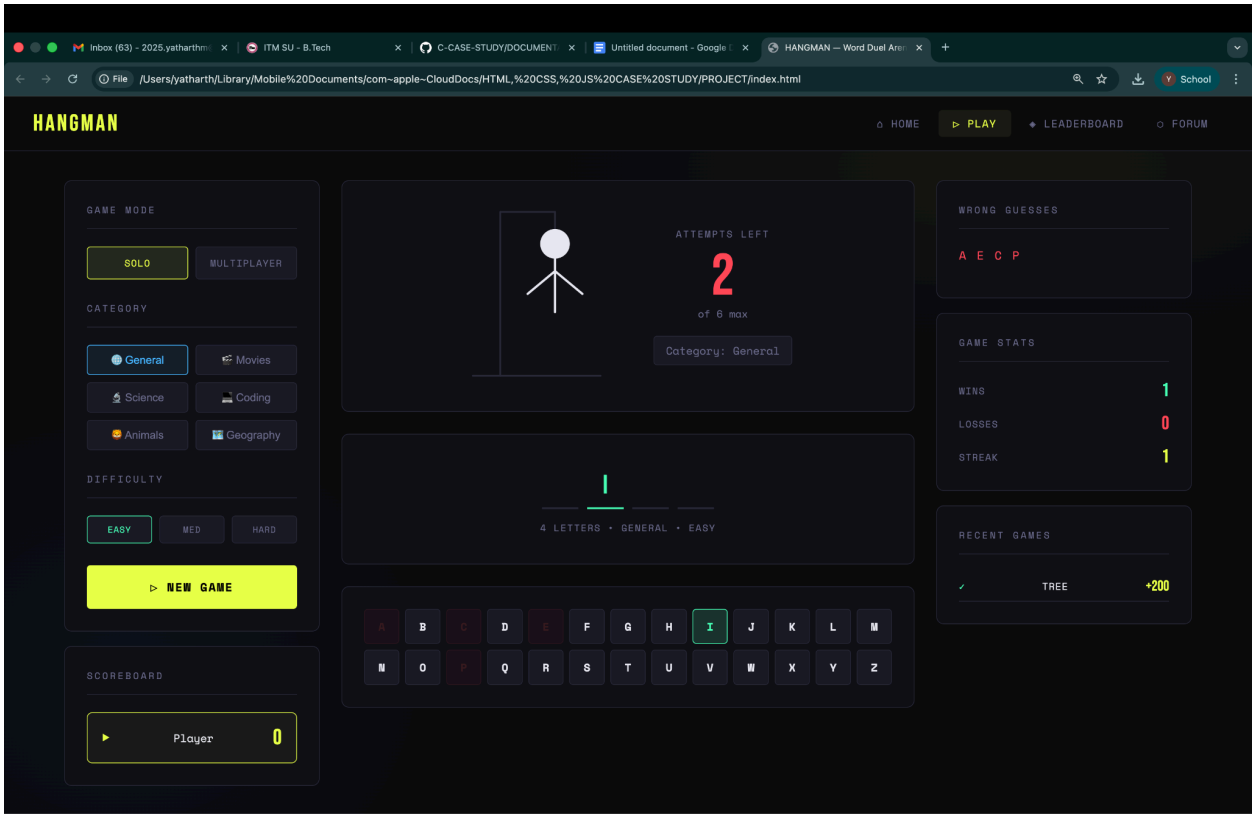
2. Gameplay interface showing hidden word and alphabet buttons

3. Hangman drawing progression during incorrect guesses

4. Leaderboard display showing player scores

5. Forum section displaying user comments

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HANGMAN
HOME
PLAY
LEADERBOARD
FORUM
GAME MODE
SOLO
MULTIPLAYER
CATEGORY
General
Movies
Science
Coding
Animals
Geography
DIFFICULTY
EASY
MED
HARD
NEW GAME
SCOREBOARD
Player
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ATTEMPTS LEFT
2
of 6 max
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N O P Q R S T U V W X Y Z
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A E C P
GAME STATS
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LOSSES 0
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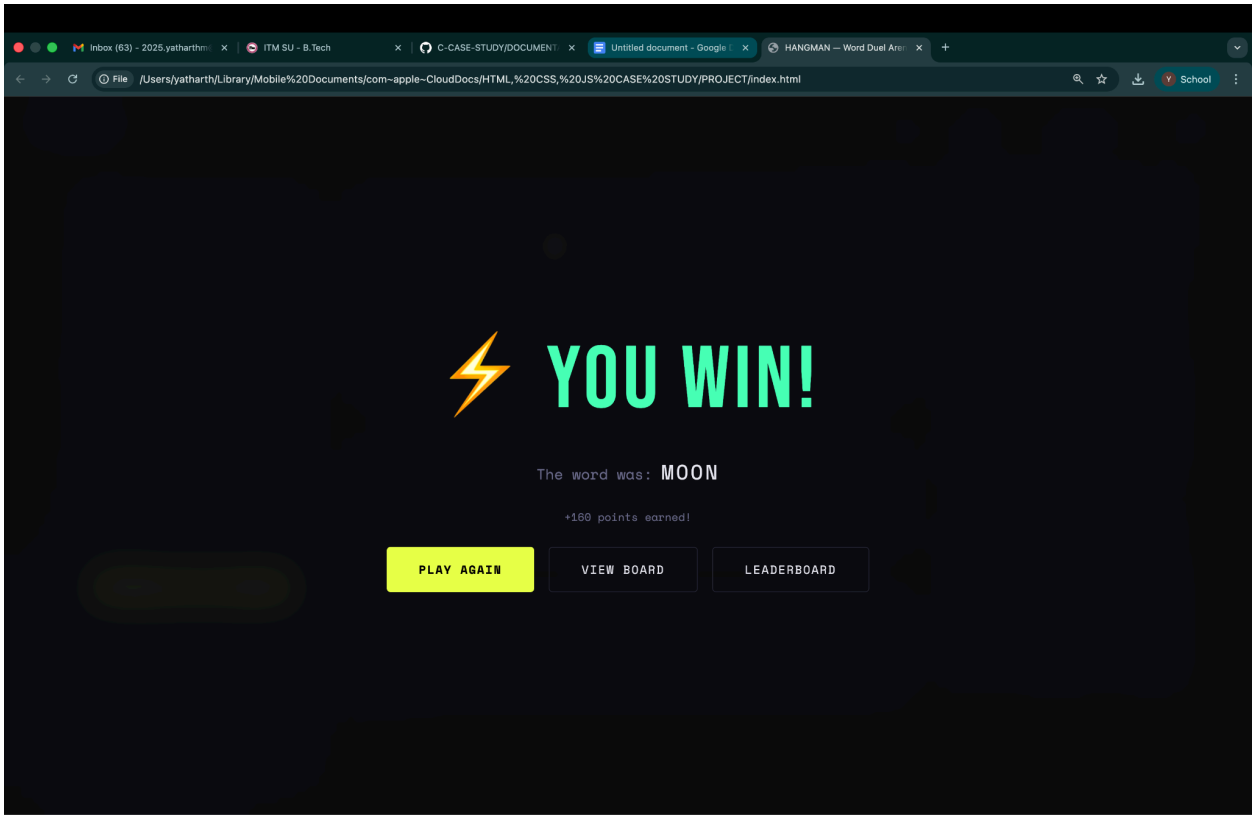
RECENT GAMES
TREE +200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File
/Users/yatharth/Library/Mobile%20Documents/com~apple~CloudDocs/HTML,%20CSS,%20JS%20CASE%20STUDY/PROJECT/index.html
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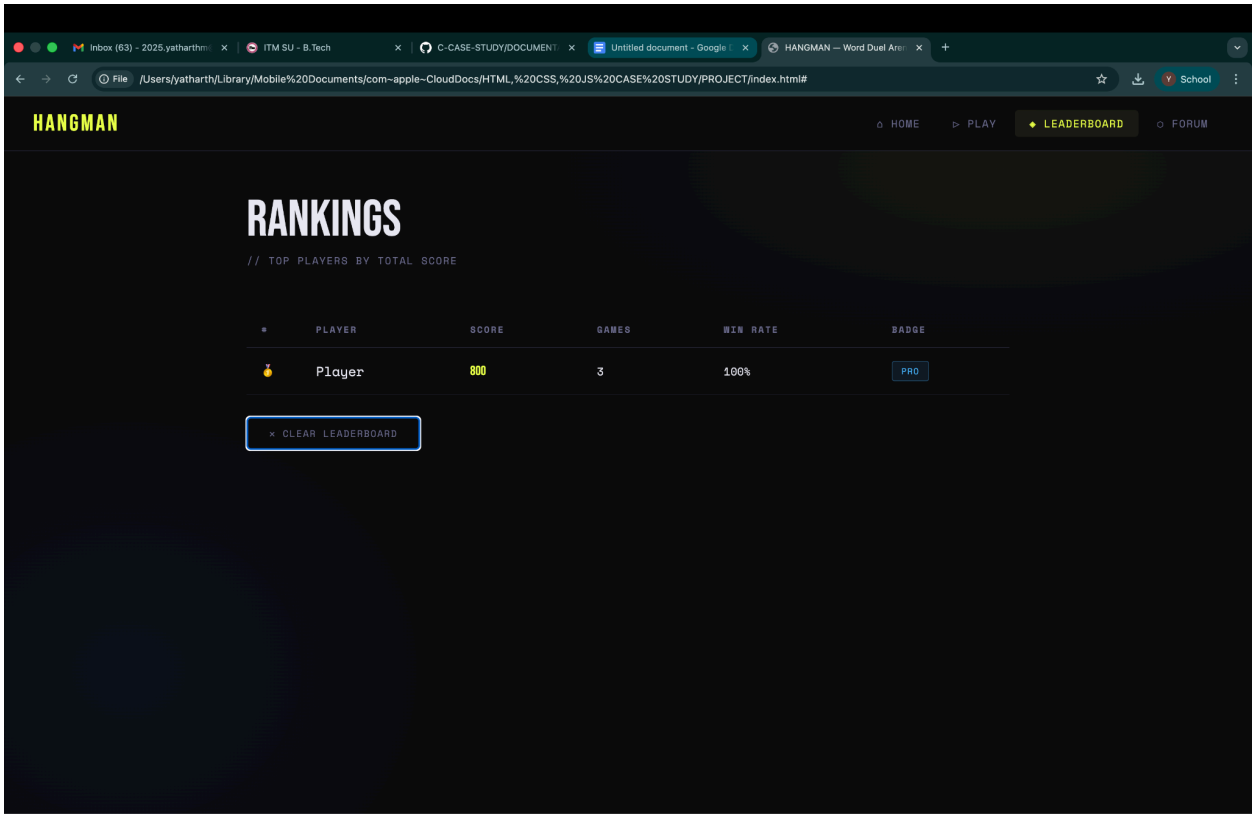
YOU WIN!
The word was: MOON
+160 points earned!
PLAY AGAIN
VIEW BOARD
LEADERBOARD



These snapshots help illustrate the functionality and user interface of the application.

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6. Problem Statement / Case Study Results and Conclusion

The Hangman Multiplayer platform successfully demonstrates how a classic word guessing game can be enhanced using modern web technologies. The application provides an interactive environment where users can play the game while tracking their progress through a leaderboard system.

The multiplayer functionality allows players to compete locally, which increases the entertainment value and encourages repeated gameplay. The inclusion of a discussion forum also allows users to share strategies and interact with other players.

The use of HTML, CSS, and JavaScript ensures that the application is lightweight and accessible through any modern web browser. Since the platform operates entirely on the frontend, it does not require a server or database for operation. Local storage is used to maintain user scores and forum comments within the browser.

Overall, the project successfully achieves its objective of creating an engaging and responsive Hangman game platform while demonstrating the practical application of frontend web development concepts.

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7. References

1. Mozilla Developer Network Web Docs for HTML CSS and JavaScript documentation

2. W3Schools tutorials for web development concepts

3. Figma design tool for user interface prototyping

4. Various online resources explaining Hangman game logic and word guessing algorithms

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